Nipah, Zika, and Monkeypox

131

Nipah Virus

Etiology and Epidemiology

- ☑ Nipah is an enveloped ribonucleic acid (RNA) virus belonging to the Paramyxoviridae family and genus *Henipavirus*. It is a zoonosis with fruit bats as the natural hosts and pigs as intermediate hosts.
- ☑ The virus is transmitted from animals to humans through direct contact or consumption of food contaminated by the animals and from humans to humans through close contact or inhalation of contaminated aerosols.
- ☑ The virus was first identified in Nipah, Malaysia in 1998–1999 in cases of febrile encephalitis in abattoir workers due to contact with infected pigs. Since 2001, many outbreaks have been reported in Bangladesh attributed to consumption of raw date palm juice contaminated by excreta of fruits bats. Two lineages have emerged—Malaysian and the Bangladesh lineage which differ in transmissibility, clinical features and severity.
- ☑ Outbreaks in India due to the Bangladesh lineage of the virus were first reported in West Bengal in 2001 and again in 2007. The recent outbreak was in Kerala in 2018 with 18 confirmed cases and 17 deaths; the index case was a young male with a possible exposure to bats and all other cases were due to secondary transmission. In 2021, death of a 12-year-old boy due to nipah virus (NiV) due to ingestion of contaminated fruit was reported from Kerala.

Clinical Features

- ☑ The incubation period is variable but usually from 6 to 14 days. Severity varies from a mild illness to rapidly fatal encephalitis.
- ☑ The usual symptoms include fever, fatigue, nausea, vomiting, and cough progressing to breathlessness, drowsiness, seizures, and coma. Respiratory symptoms/complications are more common with the Bangladesh lineage. Relapse of encephalitis months to years after infection has also been reported.
- ☑ The case fatality ranges from 70 to 100%.

Nipah Virus

Diagnosis

- ☑ The common clinical differentials include Japanese encephalitis, dengue, chikungunya, rickettsial disease, malaria, leptospirosis, rabies, and bacterial meningitis.
- ☑ The complete blood count (CBC) shows leukopenia and thrombocytopenia. The MRI shows multifocal discrete lesions in the subcortical white matter.
- ☑ Diagnosis in early disease can be achieved by reverse transcription polymerase chain reaction (RT-PCR)/viral isolation in nasopharyngeal/oropharyngeal swabs, blood, cerebrospinal fluid (CSF), and urine. In later stages, diagnosis is possible by testing for specific immunoglobulin M (IgM) antibodies in serum. Samples can be sent to National Institute of Virology, Pune.

Treatment

- ☑ Treatment is largely symptomatic and supportive.
- ☑ Ribavirin was used in the Malaysian outbreak in a nonrandomized fashion with a 36% reduction in mortality. Other antivirals including remdesivir and favipiravir may have activity against NiV. The monoclonal antibody m102.4 has shown to be effective in protecting African green monkeys against lethal NiV challenge.

Prevention

- ☑ Prevention strategies include avoidance of raw date palm juice, partially eaten fruits, fruits found on the ground and exposure to sick bats/pigs, and areas where bats roost.
- ☑ NiV suspect patients should be quickly isolated with droplet, contact, and airborne precautions. Caregivers/healthcare workers should wear personal protective equipment with masks, goggles, gloves, and gowns and practice good hand hygiene.

Zika Virus

Etiology and Epidemiology

- ☑ This is an RNA *Flavivirus* closely related to dengue, yellow fever, and Japanese encephalitis (JE) virus and first identified in a rhesus monkey in the Zika forest of Uganda in 1947.
- ☑ It is transmitted by *Aedes aegypti*/*Aedes albopictus* and also through the sexual and perinatal route.
- ☑ Large outbreaks were reported in 2007 in Micronesia and in 2013–2014 in French Polynesia and New Caledonia. The largest outbreak till date began in Brazil in 2014 and spread to other nations in the American continent, peaked in 2016 and declined thereafter. In 2022, 31,000 cases have been reported of which 8.5% are laboratory confirmed.
- ☑ Zika cases have been reported in India since 2017. Outbreaks in Kerala, Uttar Pradesh, and Maharashtra have been reported in 2021 amidst the coronavirus disease-2019 (COVID-19) pandemic.

Clinical Features

- ☑ It is an acute illness characterized by low-grade fever, rash, arthralgia and myalgia, conjunctivitis, and lymphadenopathy.
- ☑ The disease mimics dengue/chikungunya like illness. Differentiating features are listed in **Table 1**.

TABLE 1: Differentiating features of Zika, dengue and chikungunya.

	Zika	***Dengue***	***Chikungunya***
Fever	++	+++	+++
Rash	+++	+	++
Conjunctivitis	++	–	–
Arthralgia	++	+	+++
Myalgia	+	++	+
Headache	+	++	++
Hemorrhage	–	++	–
Shock	–	+	–

Zika Virus

Complications

- ☑ Zika has been causally linked with Guillain–Barré syndrome.
- ☑ The greatest risk with Zika is mother-to-child transmission (MTCT) and congenital infection. MTCT rates are around 30% and can occur even with asymptomatic illness.
- ☑ Consequences include cranial/brain abnormalities including microcephaly/intra-ventricular calcification (1–6%), ocular abnormalities (microphthalmia, coloboma, cataract, and retinal changes), limb abnormalities (arthrogryposis multiplex congenital), and neurologic sequelae (hearing loss, seizures, and developmental issues).
- ☑ Fetal loss, preterm delivery, intrauterine growth restriction (IUGR) and low birth weight (LBW) is reported in 4–7% of affected pregnancies.

Diagnosis

- ☑ The definitive diagnosis in the first week is by RT-PCR in the blood or urine. The urine and semen may be PCR positive even up to 30–60 days after onset of illness.
- ☑ Diagnosis in later phase of the illness or in PCR negative patients can be achieved by IgM enzyme-linked immunosorbent assay (ELISA). The Zika antibodies can cross react with dengue and chikungunya and positive results should be confirmed by plaque reduction neutralization assays.

Treatment

- ☑ Treatment is supportive and symptomatic.
- ☑ Prevention entails vector control, avoidance of travel to Zika endemic areas, especially by pregnant women, and personal protection measures to prevent mosquito bites.

Monkeypox

Etiology and Epidemiology

- ☑ This is a deoxyribonucleic acid (DNA) virus of the Poxviridae family closely related to smallpox, vaccinia, and cowpox. Smallpox vaccination gives partial protection against monkeypox.
- ☑ The natural hosts are animals including monkeys, rats, squirrels, and other primates. The virus spreads from animals to humans/humans to humans through inhalation of large droplets, direct contact, and through contaminated fomites.
- ☑ The first human case was reported in 1970 in Democratic Republic of Congo (DRC). The virus is currently endemic in Central Africa (Clade 1) and Western Africa (Clade 2). There is a steady increase in cases in Africa through the years due to waning immunity after stoppage of small pox vaccination in 1980s.
- ☑ Outbreaks outside Africa were infrequent till 2022. Since May 2022, a virus belonging to clade 2 has caused >80,000 cases and 55 deaths in 111 countries. The current epidemic has mainly affected male homosexuals. India has reported 17 cases and 1 death till date.

Monkeypox

Clinical Features

- ☑ *Incubation period:* 10–14 days.
- ☑ The prodromal phase (1–4 days) is characterized by fever, body ache, fatigue, headache, sore throat, and lymphadenopathy.
- ☑ The rash appears first on the face and hands and legs and spreads inward. Private parts, palms, and soles are involved. The number of lesions vary from 1 to 500. The rash passes through macular, papular, vesicular, and then pustular phase, and lasts for 2–4 weeks. The lesions are intensely painful till they crust when they become itchy. The lesions heal with scarring.
- ☑ Complications include secondary skin infections, encephalitis, pneumonia, ocular complications leading to blindness, and scars. Infection of pregnant women may lead to abortions and stillbirth. Complications are higher in children, pregnant women, the immunocompromised, and those who have not received smallpox vaccine.

Diagnosis and Differential Diagnosis

- ☑ The diagnosis may be confirmed by PCR on lesions, nasopharyngeal swab, blood, and urine. Samples can be sent to National Institute of Virology, Pune.
- ☑ It can be differentiated from chickenpox by a shorter incubation period, centripetal spread of rash, absence of lymphadenopathy, and lesions on palms and soles in the latter. The other differentials are hand, foot, and mouth disease; secondary syphilis; insect bite allergy; molluscum contagiosum, and impetigo.

Treatment

- ☑ It is supportive and symptomatic and includes fluids, antipyretics, and analgesia for pain.
- ☑ Tecovirimat (ST-246) is a specific antiviral and is approved in USA and Europe.
- ☑ Vaccinia immunoglobulin (obtained from serum of people vaccinated for smallpox) and used to treat postvaccination progressive smallpox disease may be used in immunocompromised in conjunction with antivirals.

Prevention

- ☑ Those infected have to be isolated in single rooms. Contact and droplet precautions have to be followed till all scabs have fallen off.
- ☑ Smallpox vaccination gives partial protection against monkeypox. Vaccination is recommended for high risk population, healthcare workers, and for postexposure prophylaxis. ACAM2000 and modified vaccinia Ankara are the currently available vaccines.

Further Reading

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Encephalitis in Children

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Encephalitis in Children

132

Introduction

- ☑ Encephalitis (inflammation of the brain parenchyma), should be considered in any child who presents with altered sensorium for ≥24 hours and at least one of the following features—fever, seizures, and focal neurological findings.
- ☑ Often, there are associated laboratory evidences of cerebrospinal fluid (CSF) pleocytosis, focal/generalized electroencephalogram (EEG) abnormalities, or abnormal neuroimaging. Differentiate it from encephalopathy and meningitis:
  - *Encephalopathy:* Brain dysfunction in the absence of inflammation, and caused by metabolic, hypoxic, ischemic, or toxic states
  - *Meningitis:* Inflammation of the meninges with the absence of abnormal brain function, clinically manifesting as fever, headache, vomiting, photophobia, and meningeal signs.
- ☑ Encephalitis can be infectious or noninfectious inflammation. Most common infectious agents are viruses whereas noninfectious causes include immune-related, demyelination, etc. Viruses can be sporadic such as herpes simplex or epidemic such as Japanese encephalitis (JE). Other viral infections such as measles, mumps, varicella-zoster, dengue, chikungunya, human immunodeficiency virus (HIV), rabies, and enteroviral infections can also have encephalitis as a complication.
- ☑ *Acute encephalitis syndrome (AES)* is a term used by the WHO for syndromic surveillance in the context of JE and is defined as a person of any age, at any time of the year with acute onset of fever and a change in mental status (symptoms as confusion, disorientation, coma, or inability to talk) and/or new onset of seizures. JE can be either laboratory confirmed or probable when a suspected case with these features of AES is found in close geographic and temporal relationship to a laboratory confirmed case of JE in the context of an outbreak.

Presentation

The clinical picture consists of two phases.

1. *Prodromal phase:* Lasts 1–3 days prior, with fever, malaise, and headache
2. *Encephalitic phase:* Persistent fever, decreasing level of consciousness, seizures, abnormal movements, or focal neurological deficits

Depending on the severity and age of the child and immune responses, some may succumb and some recover through postencephalitic sequelae-cognitive decline, paresis, involuntary movements, and seizures. Some can have complete recovery.

Possible viral etiological agents based on clinical presentation are presented in **Table 1**.

TABLE 1: Possible viral etiological agents based on clinical presentation.

Focal encephalitis (frontotemporal)	HSV
Cerebellar ataxia	Varicella-zoster virus (VZV)
Dementia	HIV and measles virus
Extrapyramidal manifestations	JE, West Nile virus, and Nipah virus (autoimmune encephalitis)
Retinitis	CMV and West Nile virus
Rash	Measles, rubella, VZV, and hand, foot, and mouth disease
Diarrhea	*Enterovirus*
RTI features	Influenza, H1N1, and adenovirus
Parotitis	Mumps
Lymphadenopathy	HIV, EBV, CMV, rubella, West Nile virus
Hepatitis	*Coxiella burnetii*

(CMV: cytomegalovirus; EBV: Epstein–Barr virus; HIV; human immunodeficiency virus; HSV: herpes simplex virus; JE: Japanese encephalitis; RTI: respiratory tract infection)

Evaluation (Table 2)

TABLE 2: Evaluation of encephalitis.

In all children with suspected encephalitis	Complete blood count, blood culture, electrolytes/blood glucose, liver, and kidney function tests, peripheral smear for the malarial parasite, and chest X-ray
In most children	Neuroimaging, cerebrospinal fluid (CSF) analysis after stabilization if no contraindications, electroencephalogram (EEG) if seizure or suspicion of nonconvulsive status epilepticus (NCSE) or autoimmune encephalitis (AIE), extreme delta brush in AIE, and periodic lateralized epileptiform discharges (PLEDs) in herpes simplex virus (HSV)
In selected children, where initial evaluation is negative, but strong suspicion of encephalitis persists: to confirm inflammatory nature of disease, and rule out alternate causes	Autoimmune workup—antineuronal [NMDAR and voltage-gated potassium channel (VGKC)], antithyroid microsomal/thyroglobulin antibodies, antinuclear antibody (ANA), erythrocyte sedimentation rate (ESR), aquaporin-4 Ab in neuromyelitis optica (NMO) and myelin oligodendrocyte glycoprotein (MOG IgG) in serum. Inborn errors of metabolism (IEM) screen—arterial blood gas (ABG), plasma lactate, ammonia, urine ketones, dried blood spot for tandem mass spectrometry (TMS), urine gas chromatography–mass spectrometry (GCMS)—persistent hypoglycemia in an young infant/child consider fatty acid oxidation disorders